

STA201 Assignment 5 Solution

1. A rental car company claims the mean time to rent a car on their website is 60 seconds with a standard deviation of 22 seconds. A random sample of 36 customers attempted to rent a car on the website. The mean time to rent was 55 seconds. Is this enough evidence to say that the mean time is less than the company's claim? Consider $\alpha=0.05$.

Solution:

$$H_0 : \mu = 60$$

$$H_1 : \mu < 60$$

$$\mu_0 = 60, \sigma = 22, \bar{x} = 55, n = 36$$

Therefore, this is a left-tailed Z-Test.

$$\therefore z\text{-stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = -1.36$$

Critical Region Approach

The tabulated value of z for a left-tailed test at 5% level of significance is -1.645

Hence, C.R. < -1.645

The calculated value of z-stat = -1.36 > -1.645, which is not within the critical region. Therefore, we fail to reject the null hypothesis. There is not significant evidence to conclude that the true mean is less than the company's claim.

P-value approach

$$P\text{-value} = P(z < z\text{-stat}) = P(z < -1.36) = 0.0869$$

$$\text{So, } p\text{-value} > \alpha \Rightarrow 0.0869 > 0.05$$

Therefore, we fail to reject the null hypothesis. There is not significant evidence to conclude that the true mean is less than the company's claim.

2. A teacher wanted to know how well the gifted students in her class perform relative to her other classes. She administers a standardized test with a mean of 50 and standard deviation of 10. Her class of 31 students has an average score of 55. Test the claim that their average score is higher than 50? Consider $\alpha=0.05$.

Solution:

$$H_0 : \mu = 50$$

$$H_1 : \mu > 50$$

$$\mu_0 = 50, \sigma = 10, \bar{x} = 55, n = 31$$

Therefore, this is a right-tailed Z-Test.

$$\therefore z\text{-stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = 2.78$$

Critical Region Approach

The tabulated value of z for a right-tailed test at 5% level of significance is 1.645

Hence, C.R. > 1.645

The calculated value of z-stat = 2.78 > 1.645, which is within the critical region. Therefore, we are able to reject the null hypothesis. It is statistically significant that the class average is higher than 50 at 5% level of significance.

P-value approach

$$P\text{-value} = P(z > z\text{-stat}) = P(z > 2.78) = 1 - P(z < 2.78) = 0.0027$$

$$\text{So, } p\text{-value} < \alpha \Rightarrow 0.0027 < 0.05$$

Therefore, we are able to reject the null hypothesis. It is statistically significant that the class average is higher than 50 at 5% level of significance.

3. A researcher wanted to study the effects of mentoring on intelligence scores. He wanted to know as a baseline what the average intelligence of his students were relative to the general population. He used a standardized IQ test which has a mean of 100 and standard deviation of 16. The 50 students in his study scored an average of 102 on the IQ test. Does his study support the baseline? Consider $\alpha=0.05$.

Solution:

$$H_0 : \mu = 100$$

$$H_1 : \mu \neq 100$$

$$\mu_0 = 100, \sigma = 16, \bar{x} = 102, n = 50$$

Therefore, this is a two-tailed Z-Test.

$$\therefore z\text{-stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = 0.88$$

Critical Region Approach

The tabulated value of z for a two-tailed test at 5% level of significance is 1.96

Hence, C.R. > 1.96

The calculated value of z-stat = 0.88 < 1.96, which is not within the critical region. Therefore, we fail to reject the null hypothesis. We conclude at 5% level of significance that the mean performance of the class is the same as the baseline.

P-value approach

$$P\text{-value} = 2 * P(z > z\text{-stat}) = 2 * P(z > 0.88) = 2 * [1 - P(z < 0.88)] = 0.3788$$

$$\text{So, } p\text{-value} > \alpha \Rightarrow 0.3788 > 0.05$$

Therefore, we fail to reject the null hypothesis. We conclude at 5% level of significance that the mean performance of the class is the same as the baseline.

4. Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that mean burning rate must be 50 centimeters per second. The experimenter selects a random sample of 25 propellants and obtains a sample average burning rate of 48.7 centimeters per second with standard deviation of 2 centimeters per second. What conclusion can be drawn from the samples?

Solution:

$$H_0 : \mu = 50$$

$$H_1 : \mu \neq 50$$

$$\mu_0 = 50, n = 25, \bar{x} = 48.7, s = 2$$

Therefore, this is a two-tailed T-Test

$$\therefore t\text{-stat} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = -3.25$$

$$df = n - 1 = 25 - 1 = 24$$

$$\text{Let, } \alpha = 0.05$$

Critical Region Approach

The tabulated value of t for a two-tailed test at 5% level of significance with 24 degrees of freedom is 2.064

Hence, C.R. > 2.064

The calculated value of $|t\text{-stat}| = 3.25 > 2.064$, which is within the critical region. Therefore, we are able to reject the null hypothesis. From the samples, it can be concluded at 5% level of significance that the pipes do meet specifications.

P-value approach

$P\text{-value} = 2 * P(t < t\text{-stat}) = 2 * P(t < -3.25) = 0.0034$

So, $p\text{-value} < \alpha \Rightarrow 0.0034 < 0.05$

Therefore, we are able to reject the null hypothesis. From the samples, it can be concluded at 5% level of significance that the pipes do meet specifications.

5. You have just taken ownership of a pizza shop. The previous owner told you that you would save money if you bought the mozzarella cheese in a 4.5 pound slab. Each time you purchase a slab of cheese, you weigh it to ensure that you are receiving 72 ounces of cheese. The results of 7 random measurements are 70, 69, 73, 68, 71, 69 and 71 ounces. Are these differences due to chance or is the distributor giving you less cheese than you deserve?

Solution:

$$H_0 : \mu = 72$$

$$H_1 : \mu < 72$$

$$\mu_0 = 72, n = 7, \bar{x} = \frac{(70+69+73+68+71+69+71)}{7} = \frac{491}{7}, s = 1.68$$

Therefore, this is a left-tailed T-Test

$$\therefore t\text{-stat} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = -2.92$$

$$df = n - 1 = 7 - 1 = 6$$

Let, $\alpha = 0.05$

Critical Region Approach

The tabulated value of t for a left-tailed test at 5% level of significance with 6 degrees of freedom is 1.943

Hence, C.R. < -1.943

The calculated value of $t\text{-stat} = -2.92 < -1.943$, which is within the critical region. Therefore, we are able to reject the null hypothesis. From the 7 random measurements, it is statistically significant to conclude, at 5% level of significance, that the distributor is giving less cheese than deserved.

P-value approach

$P\text{-value} = P(t < t\text{-stat}) = P(t < -2.92) = 0.0133$

So, $p\text{-value} < \alpha \Rightarrow 0.0133 < 0.05$

Therefore, we are able to reject the null hypothesis. From the 7 random measurements, it is statistically significant to conclude, at 5% level of significance, that the distributor is giving less cheese than deserved.